

Equivalência de Domínios — UStAI ↔ Synthetic User Stories

Projeto IN-0953-ReFair · Correspondência entre o eixo de domínio do dataset UStAI-annotated_V2 e os eixos de domínio de aplicação e ML task do dataset Synthetic User Stories (ReFAIR).

1. Objetivo

Mapear cada user story do dataset **UStAI** (1260 user stories) para o vocabulário do dataset **Synthetic User Stories** do ReFAIR, produzindo um dataset de equivalência reutilizável.

2. Os dois datasets

Dataset	Coluna usada como domínio	Cardinalidade
UStAI (UStAI-annotated_V2.pdf)	role_shorten (papel da user story)	1260 US · 179 papéis · 42 abstracts
Synthetic User Stories (.xlsx)	Domain (domínio de aplicação)	34 domínios
Synthetic User Stories (.xlsx)	Machine Learning Task	367 ML tasks

Ressalva: o PDF Synthetic-User-Stories.pdf entregue contém a lista das 367 ML Tasks, não a coluna de domínio. A coluna Domain real tem 34 domínios de aplicação. Por isso a equivalência foi feita para os dois eixos: domínio de aplicação e ML task.

3. Metodologia

O **role_shorten** é a âncora do lado UStAI. Como cada **abstract** (A1–A42) descreve um único sistema de IA, as 30 user stories de um abstract compartilham o mesmo domínio e a mesma ML task; papéis genéricos (researcher, developer, data scientist...) aparecem em dezenas de abstracts e não determinam domínio sozinhos — logo o domínio é fixado por abstract (desambiguação pelo contexto). Cada abstract recebe ainda um nível de confiança (High/Medium/Low) para o mapeamento de domínio.

4. Tabela de equivalência por abstract

Cada abstract = 30 user stories (10 Gemini_1.5_flash + 10 Llama 3.1 70b + 10 O1_mini).

Abs.	Tema do sistema de IA	Domínio equiv.	ML Task equiv.	Conf.
A1	Autonomous vehicles (AI + real-time data for perception & collision avoidance)	Transportation	machine perception	High
A2	Wearable-based detection of physical aggression in children	Psychology	time-series classification	Medium
A3	Real-time aggression detection on Twitter	Social Media	text classification	High
A4	AI-powered military drones & international security	Political Science	reinforcement learning	Low
A5	GoCart autonomous delivery robot in healthcare/residential facilities	Health	robot learning	High
A6	ADAS driver-intention recognition with Gaussian Mixture Models	Transportation	mixture model	High
A7	ML-based mental health / mental illness prediction	Psychology	classification method	Medium
A8	Anomaly detection in smart-health wearable data	Health	anomaly detection	High
A9	Heart disease diagnosis with Fuzzy SVM & incremental learning	Cardiology	support vector machine	High
A10	Blood-glucose / hypoglycemia prediction with Support Vector Regression	Endocrinology	support vector regression	High
A11	Anti-counterfeiting copy-detection patterns (digital twin, image-to-image GANs)	Computer Vision	generative adversarial network	Medium
A12	Firearm Violence Vulnerability Index (XGBoost community-risk forecasting)	Social Work	gradient boosting	Low
A13	AI-DrugNet drug repurposing for Alzheimer's disease	Pharmacology	deep learning	High
A14	NYC taxi / Uber ridership prediction	Transportation	forecasting algorithms	High
A15	Income / poverty prediction from demographic & lifestyle data	Economics	classification method	High

A16	Newborn cry diagnostic system (cry-audio classification)	Pediatrics	classification method	High
A17	Sleep apnea / hypopnea prediction with LAMSTAR neural networks	Medicine	artificial neural network	High
A18	Tuberculosis detection from chest X-rays with CNNs	Radiology	convolutional neural network	High
A19	Credit-card default prediction	Finance & Marketing	classification method	High
A20	Drug-abuse prediction from Big Five personality traits	Psychology	classification method	Medium
A21	Racial & gender bias in online freelancing marketplaces	Economics	data mining	Low
A22	Multidimensional freelancer assessment framework for online labor	Economics	learning ranking	Low
A23	Search-engine bias analysis (PAWS platform)	Information Systems	information retrieval	Medium
A24	ArnetMiner academic search / academic social network	Library	information retrieval	High
A25	E-commerce recommendation (item-to-item collaborative filtering)	Information Systems	recommendation system	Medium
A26	Explainable recommendation justifications (Seq2Seq, aspect-planning)	Information Systems	natural language generation	Medium
A27	Session-based recommendation with RNNs	Information Systems	recommendation system	Medium
A28	Reciprocal dating recommender from speed-dating data	Social Networks	recommendation system	Medium
A29	Historical census record linkage	Demography	classification method	High
A30	Recidivism prediction with Decision Trees (criminal justice)	Law	decision tree	High
A31	Advertisement identification on web pages	Finance & Marketing	classification method	Low
A32	Stance detection with multi-task learning (NLP)	Linguistics	multi-task learning	Medium
A33	Semantic news search with Wikidata annotation	News	information retrieval	High
A34	DNN-based network intrusion detection system	Computer Networks	deep neural network	High
A35	Wholesale sales prediction with ARIMA	Finance & Marketing	time series forecasting	High
A36	Moving-vehicle classification via wireless sensor networks	Transportation	time-series classification	High
A37	Fantasy sports analytics & player selection	Sport	data mining	High
A38	AI recruitment & personality assessment	Psychology	classification method	Low
A39	Playlist success prediction	Music	classification method	High
A40	Compromised social-network account detection	Social Networks	anomaly detection	High
A41	Food image recognition with stacked global-local attention network	Computer Vision	attention mechanism	Medium
A42	Mobile vaccination-site placement optimization	Health	recommendation system	High

Notas (menor confiança):

A2 (Psychology): Alternativa: Pediatrics (populacao infantil).

A4 (Political Science): Nao ha dominio Militar/Defesa nos 34; Political Science e o mais proximo (seguranca internacional).

A7 (Psychology): Alternativa: Health.

A11 (Computer Vision): Dominio de aplicacao = seguranca de marca; Computer Vision e o mais proximo dos 34.

A12 (Social Work): Alternativas: Sociology / Political Science / Health (saude publica).

A20 (Psychology): Alternativas: Pharmacology / Social Work.

A21 (Economics): Mercado de trabalho online; alternativas: Social Networks / Sociology.

A22 (Economics): Alternativas: Social Networks / Information Systems.

A23 (Information Systems): Alternativa: Social Media.

A25 (Information Systems): Alternativa: Finance & Marketing.

A26 (Information Systems): Contexto = sistemas de recomendacao explicaveis.
A27 (Information Systems): Alternativa: Finance & Marketing.
A28 (Social Networks): Alternativa: Sociology.
A31 (Finance & Marketing): Publicidade; alternativa: Information Systems.
A32 (Linguistics): Alternativa: Social Media.
A38 (Psychology): Recrutamento/RH; nao ha dominio de RH nos 34. Alternativa: Economics.
A41 (Computer Vision): Alternativa de task: convolutional neural network.

5. Distribuição por domínio equivalente

Domínio	Nº user stories	Nº abstracts
Information Systems	120	4
Psychology	120	4
Transportation	120	4
Economics	90	3
Finance & Marketing	90	3
Health	90	3
Computer Vision	60	2
Social Networks	60	2
Cardiology	30	1
Computer Networks	30	1
Demography	30	1
Endocrinology	30	1
Law	30	1
Library	30	1
Linguistics	30	1
Medicine	30	1
Music	30	1
News	30	1
Pediatrics	30	1
Pharmacology	30	1
Political Science	30	1
Radiology	30	1
Social Media	30	1
Social Work	30	1
Sport	30	1

Confiança do mapeamento: High = 750 US · Medium = 330 US · Low = 180 US.

6. Equivalência role_shorten → domínio

Dos 179 papéis, 110 mapeiam para um único domínio e 69 são multi-domínio (genéricos).

6.1. Papéis com domínio único

role_shorten	Nº US	Domínio equivalente
archivist	1	Demography
banker	1	Finance & Marketing

bioinformatician	2	Pharmacology
candidate	3	Psychology
car manufacturer	2	Transportation
cardiologist	1	Cardiology
chef	1	Computer Vision
child	2	Psychology
client	9	Economics
commuter	3	Transportation
competitor in the search engine market	1	Information Systems
computational biologist	1	Pharmacology
content creator	2	Information Systems
content strategist	1	Information Systems
credit card issuer	1	Finance & Marketing
dating coach	1	Social Networks
defense contractor	1	Political Science
demographer	1	Demography
dietary staff member	1	Health
diplomat	1	Political Science
driver	9	Transportation
drone operator	1	Political Science
economic historian	4	Demography
endocrinologist	1	Endocrinology
entrepreneur	1	Pharmacology
environmentalist	2	Transportation
ethicist	1	Pediatrics
field commander	1	Political Science
forecasting expert	1	Finance & Marketing
freelancer	9	Economics
genealogist	2	Demography
GIS specialist	1	Political Science
grant writer	1	Social Work
health insurance provider	1	Endocrinology
Healthcare Administrator	1	Health
healthcare policy advisor	1	Medicine
historian	2	Demography
human-factors specialist	1	Transportation
innovator	1	Computer Vision
interoperability coordinator	1	Political Science
job seeker	2	Psychology
judge or magistrate	1	Law

kitchen staff member	1	Health
knowledge base specialist	1	News
lab technician	3	Health
law enforcement agency	1	Law
linguist	2	Linguistics
machine learning model	1	Psychology
member of the Pokec social network community	1	Social Networks
military commander	3	Political Science
modeler	1	Library
moderator	3	Social Media
music enthusiast	7	Music
music industry professional	2	Music
music label	1	Music
music producer	1	Music
music streamer	1	Music
musician	1	Music
neonatologist	3	Pediatrics
Network engineer	1	Transportation
network security specialist	1	Transportation
neurologist	3	Pharmacology
news agency	3	News
News editor	1	News
non-profit organization leader	1	Economics
non-proliferation expert	1	Political Science
OEM	1	Transportation
online retailer	7	Information Systems
organizer	2	Social Networks
participant	1	Transportation
passenger	3	Transportation
peacekeeper	1	Political Science
pediatrician	1	Pediatrics
person with dietary restrictions	1	Computer Vision
pharmaceutical company scientist	1	Pharmacology
pharmacist	1	Pharmacology
player	9	Sport
playlist creator	1	Music
playlist curator	1	Music
printing press operator	1	Computer Vision
public transportation official	1	Transportation
publisher	1	Library

radiologist	5	Radiology
recruiter	3	Psychology
research evaluator	1	Library
research funder	1	Library
resident	1	Health
residential care staff member	2	Health
restaurant operator	1	Computer Vision
resturant owner	1	Computer Vision
safety expert	3	Transportation
school counselor	1	Psychology
security expert	3	Computer Vision
shareholder of a wholesale company	1	Finance & Marketing
single person	2	Social Networks
sleep specialist	2	Medicine
social media moderator	1	Linguistics
social scientist studying the impact of technology on dating	1	Social Networks
sociologist studying assimilation and discrimination	1	Demography
software architect	1	Information Systems
sports fan	8	Sport
system	1	Health
taxi dispatcher	2	Transportation
taxi driver	1	Transportation
tester	1	Transportation
therapist	1	Psychology
traffic engineer	1	Transportation
trainer	1	Political Science
vaccination site coordinator	1	Health
visually impaired person	1	Transportation

6.2. Papéis multi-domínio (genéricos)

role_shorten	Nº US	Domínios equivalentes — Domínio (nº US)
researcher	162	Information Systems (15), Psychology (14), Economics (14), Transportation (13), Computer Vision (11), Health (9), Social Work (7), Pharmacology (6), Medicine (6), Library (6), Social Networks (6), Linguistics (6), Computer Networks (6), Cardiology (5), Pediatrics (5), Finance & Marketing (5), Demography (5), Social Media (4), Endocrinology (4), Radiology (4), News (4), Law (3), Political Science (2), Sport (1), Music (1)
developer	139	Information Systems (30), Computer Vision (11), Finance & Marketing (11), Transportation (10), Psychology (10), Health (9), Social Networks (6), Pediatrics (5), Medicine (5), Social Media (4), Cardiology (4), Endocrinology (4), Library (4), News (4), Computer Networks (4), Law (3), Linguistics (3), Radiology (2), Economics (2), Demography (2), Sport (2), Political Science (1), Social Work (1), Pharmacology (1), Music (1)
data scientist	81	Information Systems (9), Psychology (7), Transportation (7), Economics (7), Health (6), Finance & Marketing (5), Social Networks (4), Cardiology (3), Computer Vision (3), Pharmacology (3), Library (3), Demography (3), Law (3), Computer Networks (3), Social Media (2), Social Work (2), Pediatrics (2), Medicine (2), Linguistics (2), Music (2), Endocrinology (1), Radiology (1), News (1)

analyst	56	Finance & Marketing (9), Transportation (6), Social Networks (6), Psychology (4), Economics (4), Information Systems (4), Linguistics (4), Health (2), Computer Vision (2), Social Work (2), Demography (2), Computer Networks (2), Sport (2), Music (2), Political Science (1), Cardiology (1), Endocrinology (1), Law (1), News (1)
manager	45	Finance & Marketing (13), Transportation (8), Health (8), Psychology (4), Economics (4), Sport (2), Social Media (1), Computer Vision (1), Library (1), Information Systems (1), Linguistics (1), Computer Networks (1)
user	44	Social Networks (12), Computer Vision (8), Psychology (4), Information Systems (4), Finance & Marketing (4), Social Media (3), Health (2), Sport (2), Music (2), Cardiology (1), Economics (1), Computer Networks (1)
administrator	39	Psychology (7), Social Networks (5), Social Media (4), Cardiology (4), Library (4), Economics (3), Health (2), Endocrinology (2), Radiology (2), Law (2), Computer Networks (2), Transportation (1), Medicine (1)
policymaker	39	Economics (8), Psychology (6), Information Systems (4), Social Work (3), Pharmacology (3), Finance & Marketing (3), Law (2), Health (2), Political Science (1), Computer Vision (1), Transportation (1), Radiology (1), Library (1), Demography (1), Computer Networks (1), Sport (1)
advocate	29	Transportation (4), Finance & Marketing (4), Economics (4), Psychology (3), Pediatrics (2), Radiology (2), Law (2), Linguistics (2), Political Science (1), Health (1), Endocrinology (1), Social Work (1), Pharmacology (1), Information Systems (1)
officer	23	Law (5), Health (3), Social Media (2), Transportation (2), Finance & Marketing (2), Social Networks (2), Psychology (1), Political Science (1), Endocrinology (1), Computer Vision (1), Social Work (1), Economics (1), Computer Networks (1)
healthcare provider	21	Psychology (5), Cardiology (4), Medicine (4), Health (3), Radiology (2), Endocrinology (1), Pharmacology (1), Pediatrics (1)
customer	20	Information Systems (14), Finance & Marketing (5), Economics (1)
patient	19	Health (7), Radiology (4), Cardiology (3), Endocrinology (3), Medicine (2)
product manager	18	Information Systems (3), Social Networks (3), Finance & Marketing (3), Linguistics (3), Music (2), Pediatrics (1), Radiology (1), News (1), Computer Networks (1)
machine learning engineer	16	Computer Vision (3), Information Systems (2), Social Networks (2), Social Work (1), Pharmacology (1), Economics (1), Pediatrics (1), Radiology (1), Psychology (1), Demography (1), Linguistics (1), Computer Networks (1)
public health official	14	Psychology (4), Social Work (4), Health (3), Radiology (2), Pediatrics (1)
quality assurance specialist	13	Computer Vision (3), Health (1), Cardiology (1), Pediatrics (1), Medicine (1), Radiology (1), Economics (1), Information Systems (1), Social Networks (1), Computer Networks (1), Transportation (1)
global citizen	12	Information Systems (4), Economics (2), News (1), Transportation (1), Sport (1), Psychology (1), Music (1), Health (1)
strategist	11	Political Science (4), Information Systems (2), Transportation (1), Social Work (1), Social Networks (1), Sport (1), Music (1)
IT specialist	11	Health (4), Finance & Marketing (2), Cardiology (1), Computer Vision (1), Medicine (1), Law (1), Social Networks (1)
city planner	10	Transportation (8), Social Work (1), Health (1)
parent	10	Psychology (6), Pediatrics (2), Finance & Marketing (2)
HR specialist	10	Psychology (7), Economics (3)
journalist	9	News (5), Social Media (1), Information Systems (1), Linguistics (1), Health (1)
marketing professional	9	Information Systems (3), Finance & Marketing (2), Social Media (1), Economics (1), Medicine (1), Music (1)
healthcare professional	9	Psychology (3), Health (2), Pediatrics (2), Radiology (2)
cybersecurity expert	8	Political Science (3), Transportation (1), Social Media (1), Health (1), Computer Networks (1), Social Networks (1)
software engineer	8	Transportation (2), Linguistics (2), Endocrinology (1), Information Systems (1), Social Networks (1), Music (1)
person	8	Endocrinology (6), Social Media (1), Psychology (1)
librarian	8	Library (3), Demography (2), News (2), Information Systems (1)

engineer	7	Transportation (5), Health (1), Computer Networks (1)
regulator	7	Transportation (3), Finance & Marketing (2), Endocrinology (1), Pharmacology (1)
citizen	7	Health (2), Political Science (1), Economics (1), Law (1), News (1), Transportation (1)
instructor	7	Finance & Marketing (2), Transportation (1), Cardiology (1), Linguistics (1), Sport (1), Computer Vision (1)
UX designer	7	Information Systems (3), Transportation (1), Economics (1), Library (1), Social Networks (1)
community leader	7	Social Work (5), Health (2)
consumer	7	Information Systems (2), News (2), Transportation (1), Economics (1), Finance & Marketing (1)
social worker	7	Economics (3), Psychology (3), Law (1)
company	5	Pharmacology (2), Transportation (1), Finance & Marketing (1), Psychology (1)
mental health professional	5	Psychology (4), Social Media (1)
caregiver	5	Health (3), Endocrinology (2)
security professional	5	Computer Networks (3), Finance & Marketing (2)
teacher	4	Psychology (3), News (1)
system architect	4	Political Science (1), Computer Vision (1), Library (1), Information Systems (1)
specialist	4	Computer Vision (2), Cardiology (1), Law (1)
CEO	4	Finance & Marketing (3), Pharmacology (1)
decision-maker	4	Transportation (2), Law (1), Computer Networks (1)
financial advisor	4	Economics (2), Finance & Marketing (2)
nurse	3	Health (2), Pediatrics (1)
student	3	Psychology (1), Library (1), Demography (1)
clinical trial coordinator	3	Medicine (2), Pharmacology (1)
economist	3	Economics (2), Transportation (1)
non-profit organization	3	Economics (2), Psychology (1)
biomedical engineer	3	Medicine (2), Pediatrics (1)
psychologist	3	Psychology (2), Social Networks (1)
data engineer	3	Economics (1), Finance & Marketing (1), Computer Vision (1)
scholar	3	Demography (2), Library (1)
music streaming service provider	3	Information Systems (2), Music (1)
professional	2	Transportation (1), Psychology (1)
business owner	2	Transportation (1), Psychology (1)
executive	2	Transportation (1), Finance & Marketing (1)
system designer	2	Psychology (1), Social Media (1)
content moderator	2	Social Media (1), News (1)
technology developer	2	Political Science (1), Health (1)
logistics coordinator	2	Political Science (1), Finance & Marketing (1)

manufacturer	2	Transportation (1), Computer Vision (1)
government official	2	Economics (1), Law (1)
legal professional	2	Law (1), Psychology (1)
fact-checker	2	Linguistics (1), News (1)

7. Dataset gerado e limitações

O dataset completo (1260 linhas, 1 por user story) está em **equivalencia-ustai-synthetic.csv** e **.xlsx**, com as colunas: id, abstract, abstract_theme, us_num, llm, role_shorten, user_story, equivalent_domain, equivalent_ml_task, domain_confidence, mapping_notes.

Limitações: o domínio é constante dentro de cada abstract; 6 abstracts (A4, A12, A21, A22, A31, A38) têm baixa confiança por não haver domínio perfeito entre os 34; a ML task representa a técnica/tarefa central do estudo de origem.